Xiao Yiheng

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Education

Shanghai Jiao Tong University — Antai College of Economics and Management

- *Bachelor, double major of mathematics and economics* — *July 2023 - June 2027*

GPA: 92.5/100, Rank: 4/37

Courses: *Mathematical Analysis(99), Real Analysis(98), Complex Analysis(95), Functional Analysis(100), Ordinary Differential Equations(96), Abstract Algebra(90), Linear Algebra(97), Basic Topology(99), Algebraic Topology(98), Differential Geometry(99), Probability Theory(99), Stochastic Processes(95), Probability and Statistics(99).*

Research Experience

- **Nearly Optimal Bounds for Orthogonal Trace-Sum Maximization**

Advisor: Huikang Liu (SJTU)

- **Status**: First Author. Published in NeurIPS 2026 (poster).
- **Problem setting**: Synchronization on Stiefel manifold with independent subgaussian noise:

$$\max \quad \sum_{(i,j)\in\Omega}^{n} \operatorname{tr}(O_i^\top C_{ij} O_j) \qquad \text{s.t.} \quad O_i \in O_{d_i,r},\ i \in [n].$$

$$C_{ij} = G_i^\star G_j^{\star\top} + \sigma W_{ij}, \qquad (i,j) \in \Omega,$$

- **Motivations**: Motivated by a gap between theory and practice: Numerical experiments show that general power method (GPM) can converge linearly under $O(\sqrt{n})$ noise level, while theoretical literature only gives guarantee under $O(n^{\frac{1}{4}})$ noise
- **Contributions**:
  - Prove linear convergence of GPM and tightness of SDP relaxation under $\sqrt{n/\log n}$ noise level, closing the gap between theory and practice.
  - Extend the above results to the random observation setting, where each pairwise measurement is observed independently with probability $p \gtrsim \log n/\sqrt{n}$.
  - Prove an information theoretical min-max lower bound for synchronization on Stiefel manifold, showing that our our sufficient noise condition is optimal up to a $\sqrt{\log}$ factor.

- **A Unified Framework for Convex Low-Rank SDP**

Supervisor: Yinyu Ye (Stanford), Huikang Liu (SJTU), Dongdong Ge (SJTU)

- **Status**: Fisrt Author. Under review at Mathematical Programming(arXiv:2605.17883).
- **Problem setting**: Linear SDP: $\max\{\langle C, X\rangle \mid \mathcal{A}X = b, X \in \mathbb{S}_+^n\}$
- **Motivations**:
  - **Necessity of low rank:** For large scale SDP, full eigendecomposition and PSD projection are infeasible. However, low rank structure of the optimal solution can be used to alleviate the computational burden.
  - **Limitation of non-convex low rank algorithms (eg. Burer–Monteiro Method):** Rely on non-convex factorization $X = UU^\top$ and have difficulty in ensuring global optimality.
  - **Limitation of existing convex low rank algorithms (eg. CGAL, spectral bundle):**
    - **CGAL** computes only one leading eigenvector of the dual slack matrix $S(y_k) = C - \mathcal{A}^\star y_k$ at each iteration and only achieves sublinear convergence, which is too slow for acquiring high accuracy solution.
    - **Spectral bundle method** computes the top-r ($r \ll n$) eigenframe $V_k \in \mathrm{st(n,r)}$ of $S(y_k)$ at each iteration and combines these eigenvectors together with a PSD weight matrix $W_k \in \mathbb{S}_+^r$ to form a decent direction. Although it achieves local linear convergence, it requires solving smaller quadratic SDP subproblem at each iteration to determine weight W_k, which turns out to be the bottleneck.
- **Algorithm Design**:
  - Design two new spectral bundle type algorithms by substituting the PSD weight matrix W_k in classical spectral bundle with **diagnal** matrix and **diagnal dominant** matrix respectively.
  - Determine these two kinds of weight matrices by solving convex QP with simplex constraints, which are much easier to handle than quadratic SDP in practice.
  - Design a two phase strategy that uses the cheap diagonal algorithm for initialization and the diagonal dominant algorithm for local acceleration.
- **Theoretical Analysis**:
  - Develop a unified theoretical perspective to describe and analyze convex low rank SDP algorithms, including the two new algorithms proposed by us.
  - Prove global sublinear convergence for our first algorithm with diagonal weight and local linear convergence for our second algorithm with diagonal dominant weight.
  - Prove that the iteration sequence produced by the diagnol dominant algorithm is identical to that produced by classical spectral bundle when initialized sufficiently close to the optimal solution, meaning that we reduced single step complexity with no cost!

- **Numerical Experiments**: Our two new algorithms show significant acceleration on SOS and maxcut benchmarks compared with cuLoRADs and mosek

- **An Online Spectral Bundle Method for Large Scale SDP with Local Linear Convergence**
  Advisor: Yinyu Ye (Stanford), Huikang Liu (SJTU), Dongdong Ge (SJTU)
  - **Status**: First Author. Preprint (arXiv:2605.17883).
  - **Problem Setting and Motivations**: Same as the above project: A Unified Framework for Convex Low-Rank SDP
  - **Algorithm Design**: Classical spectral bundle computes eigenframe V_k and combination weight W_k exactly at each iteration, which is expensive. We maintain the tuple (V_k, W_k) from previous iteration. In each iteration, we only perform one gradient descent step for the quadratic SDP subproblem to update W_k and one Riemannian gradient descent step for the ky-fan maximization problem to update V_k. We then use (V_k, W_k) to update dual variable y_k in the same way as classical spectral bundle.
  - **Theoretical Analysis**: We proved local linear convergence of our algorithm under standard SDP regularity assumptions.

- **On the Convergence of Doubly Stochastic Primal-Dual Hybrid Gradient Method**
  Advisor: Huikang Liu (SJTU)
  - **Status**: First Author. Under review at SIAM Journal on Optimization(arXiv:2605.17883).
  - **Problem Setting**: Convex concave saddle point problem with separable structure in both primal and dual variables:

$$\min_x \max_y \quad L(x,y) := \sum_{j=1}^{m} g_j(x_j) + \sum_{i=1}^{n}\sum_{j=1}^{m} \langle A_{ij}x_j, y_i\rangle - \sum_{i=1}^{n} f_i(y_i).$$

  - **Motivation**:
    * **Separable structure:** The above separable block structure appears in various practical problems, including convex QP, lasso regression and SVM.
    * **Advantages of randomization:** By randomly choosing several primal and dual blocks to update at each step rather than performing full updates, we can not only **reduce the computation per iteration** but also **reduce the cost of communication** between different GPUs in the distributive computing setting.
  - **Algorithm Design**: Proposed **Doubly stochastic PDHG**, enabling efficient large-scale computation via doubly randomized block-coordinate updates on both primal and dual variables.
  - **Theoretical Analysis**:
    * Established an $\mathcal{O}(1/K)$ ergodic convergence rate for the expected primal-dual gap under general convex settings.
    * Further proved linear convergence of a restarted variant under a smoothed quadratic growth condition.
  - **Numerical Experiments**: Implemented experiments on soft-margin SVM and Model Predictive Control (MPC). Demonstrated that restarted DSPDHG outperforms deterministic PDHG and one-sided stochastic variants in practical scenarios.
  - **Extension to Non Bilinear Coupling Case** : Extended the above results to more general convex concave saddle point problems by replacing the linear coupling term $\langle A_{ij}x_j, y_i\rangle$ with general convex-concave function $\Phi_{ij}(x_j, y_i)$ with sufficient smoothness. The results were written in another paper(arXiv:2605.17883)

- **Online Learning of Kronecker-Structured Preconditioners**
  Advisor: Qi Deng (SJTU)
  - **Status**: First Author. Working Paper.
  - **Problem Setting**:
    * Consider matrix optimization problem: $\min_{X\in\mathbb{R}^{m\times n}} f(X) \Leftrightarrow \min_{x\in\mathbb{R}^{mn}} f(x)$
    * Solve it by gradient decent with precondition : $x_{k+1} = x_k - B_k \otimes A_k \nabla f(x_k)$, with $A_k \in \mathbb{S}^m_{++}, B_k \in \mathbb{S}^n_{++}$.
    * **Research Question:** How to design the best preconditioners $P_k = B_k \otimes A_k$?
  - **Motivations**:
    * In large scale applications such as LLM training, using unstructured preconditioner $P_k \in \mathbb{R}^{mn\times mn}$ is too expensive in both memory and computation. Using Kronecker structured preconditioner, we only need to save the two factors B_k, A_k.
    * In neural networks, the hessian matrix of the loss function can often be written as a Kronecker product of two matrices or a sum of a few Kronecker products.
  - **Algorithm Design**:
    * Design two feedback functions based on local curvature and post-step objective value respectively.
    * At each iteration, view $P_k = B_k \otimes A_k$ as decision variable and perform one online gradient descent step based on the feedback function to update B_k, A_k.
  - **Theoretical Analysis**: Assuming that f is strongly convex and sufficiently smooth, we got the following results:
    * For the algorithm with local curvature feedback, we prove that P_k will converge to the best Kronecker product approximation of the inverse hessian matrix of f at the optimal solution.
    * If the hessian matrix of f at the optimal solution is a Kronecker product of two matrices, both of our algorithms can achieve local superlinear convergence.

Honors and Awards

First Prize (Shanghai Division), Chinese Mathematics Competitions (CMC)	2024.12
Second Prize (Shanghai Division), China Undergraduate Mathematical Contest in Modeling (CUMCM)	2024.12
Silver Medal (Shanghai Division), China International College Students' Innovation Competition	2025.11
Honorable Mention, Mathematical Contest in Modeling (MCM)	2025.5
SJTU Undergraduate Scholarship (Grade B)	2025.12
First Prize in Provincial Division, Chinese Physics Olympiad (CPhO)	2022.9

Language and Skills

- **English**: IELTS: 7.5, GRE: 335 (Verbal: 165, Quantitative: 170)
- **Programming**: Proficient in implementing optimization algorithms via Python for numerical experiments, and leveraging leading commercial solvers (e.g., COPT) to solve complex optimization problems.

Research Interests

- **Continuous Optimization**: SDP, primal dual hybrid gradient method, optimization on manifolds, online preconditioning